

# Deliverable 2

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**Course:** CIS106 Linux Fundamentals

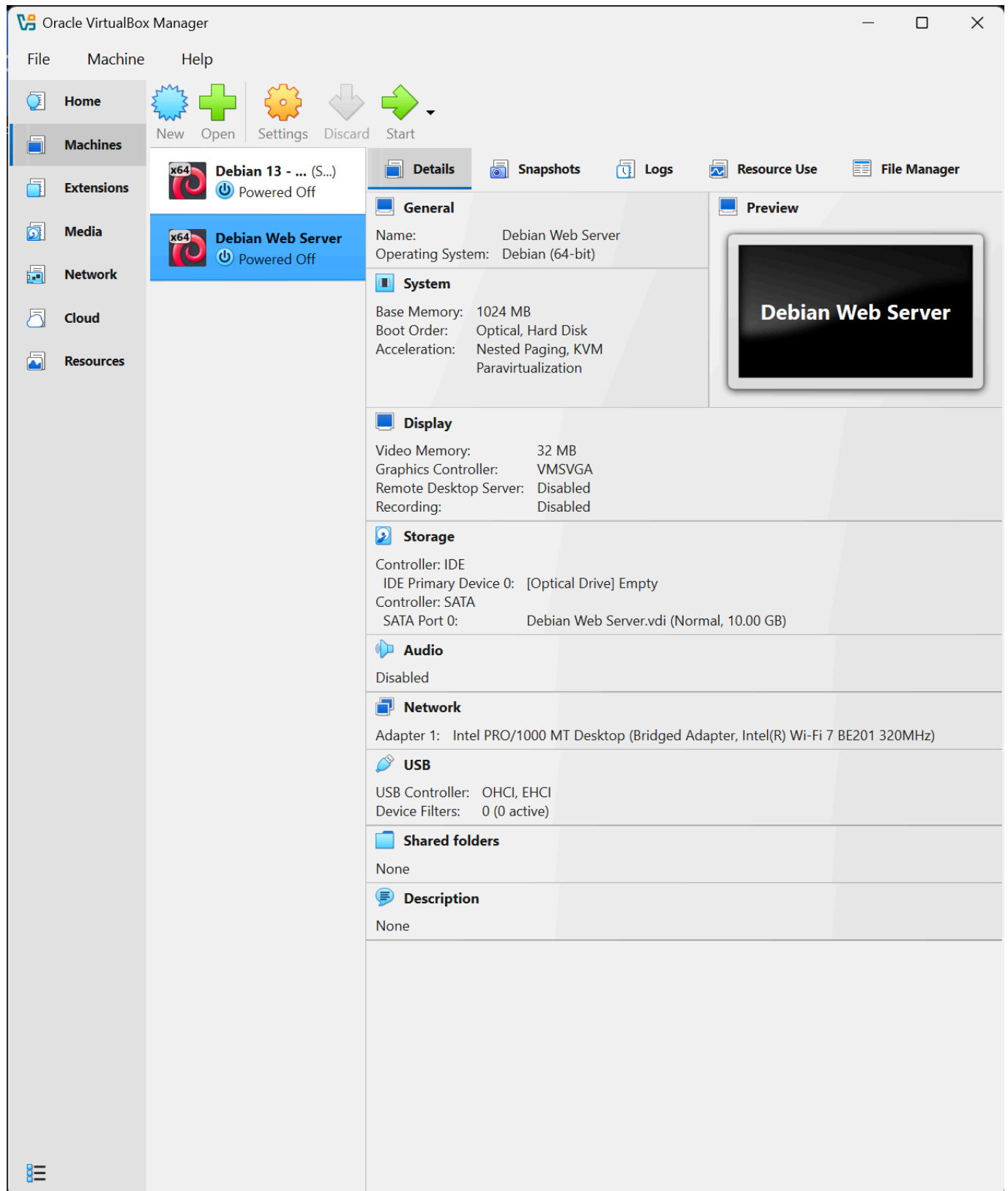
**Assignment:** Final Project - Deliverable 2

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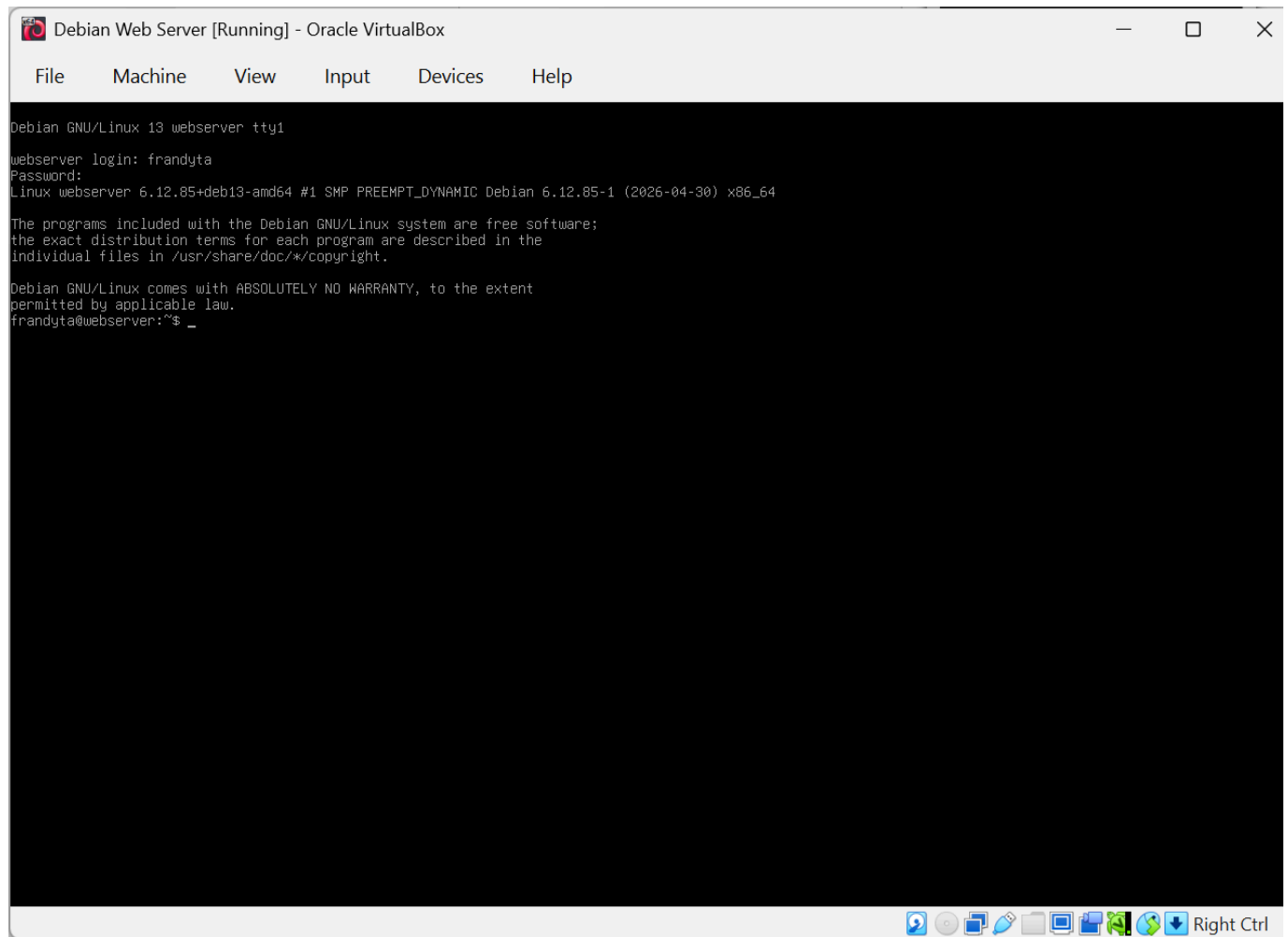
## Deliverable 2

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1. What are the server hardware specifications (virtual machine settings)?



## 2. What is the Debian Login Screen?



The screenshot shows a terminal window titled "Debian Web Server [Running] - Oracle VirtualBox". The terminal output is as follows:

```
Debian GNU/Linux 13 webserver tty1
webserver login: frandyta
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
frandyta@webserver:~$ _
```

The terminal window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". At the bottom, there is a toolbar with various icons and the text "Right Ctrl".

### 3. What is the IP address of your Debian Server Virtual Machine?

- The command used was `ip a`

```
ip a
```

- This command shows the network interfaces and IP addresses.

```

Debian GNU/Linux 13 webserver tty3
webserver login: frandyta
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

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frandyta@webserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:48:0c:65 brd ff:ff:ff:ff:ff:ff
    altname enx080027480c65
    inet 192.168.12.136/24 brd 192.168.12.255 scope global dynamic noprefixroute enp0s3
        valid_lft 604556sec preferred_lft 528956sec
    inet6 2607:fb90:2e05:65f8:a00:27ff:fe48:c65/64 scope global dynamic mngtmpaddr proto kernel_r
        valid_lft 256sec preferred_lft 256sec
    inet6 fdfe:5e6c:2e69:98de:a00:27ff:fe48:c65/64 scope global dynamic mngtmpaddr proto kernel_r
        valid_lft 3556sec preferred_lft 3556sec
    inet6 2607:fb90:2e05:65f8:5254:80af:937a:a542/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 256sec preferred_lft 256sec
    inet6 fdfe:5e6c:2e69:98de:f2f:fb3:b977:73aa/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 3556sec preferred_lft 3556sec
    inet6 fe80::b5d8:2b66:7dd6:30be/64 scope link
        valid_lft forever preferred_lft forever
frandyta@webserver:~$ hostname -I
192.168.12.136 2607:fb90:2e05:65f8:a00:27ff:fe48:c65 fdfe:5e6c:2e69:98de:a00:27ff:fe48:c65 2607:fb90:2e05:65f8:5254:80af:937a:a542 fdfe:5e6c:2e69:98de:f2f:fb3:
977:73aa
frandyta@webserver:~$ _

```

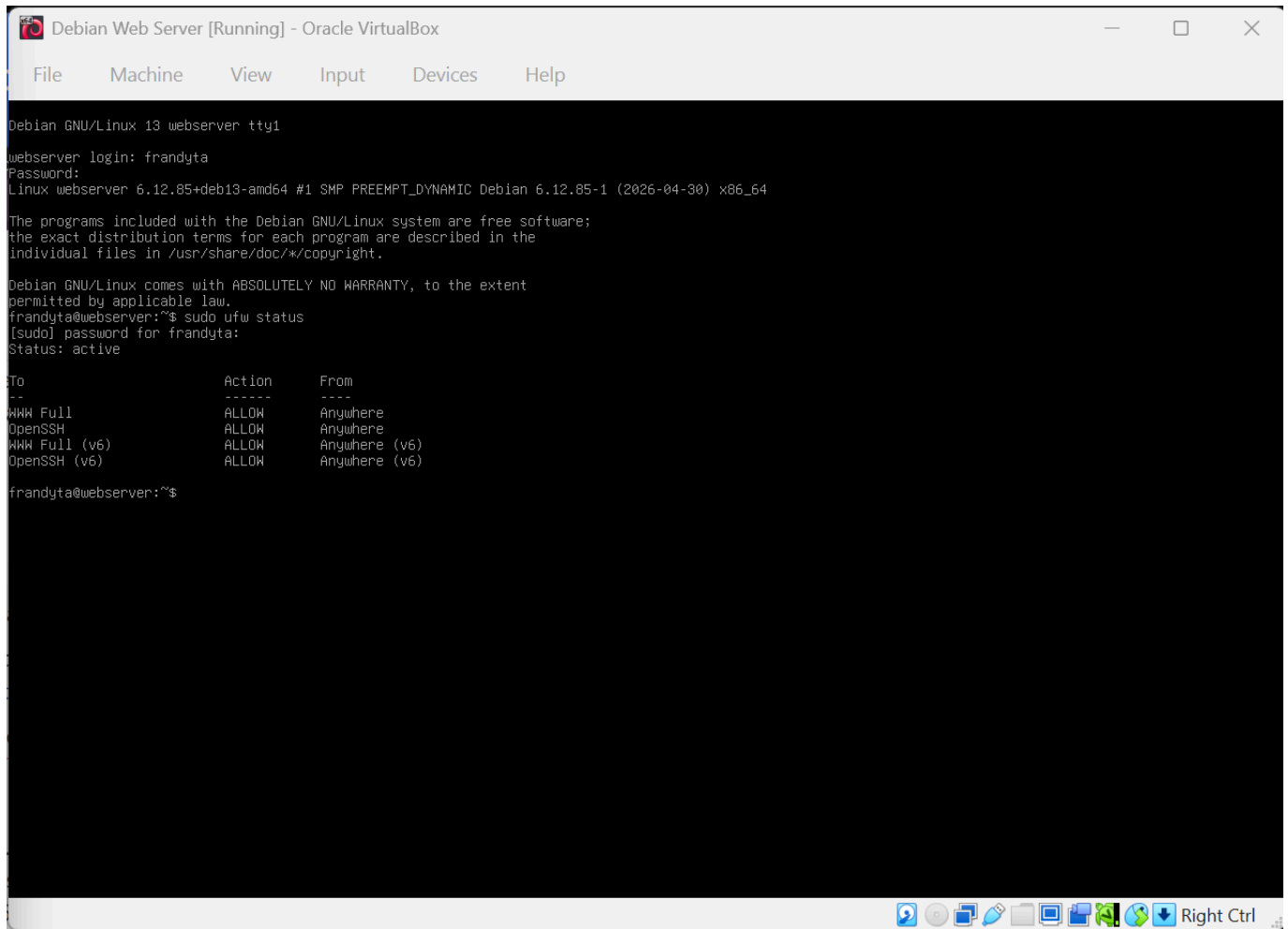
## 4. How do you work with the Firewall in Debian?

Check if the Firewall is running

- The command is:

```
sudo ufw status
```

- This command checks if the firewall is active or inactive.



```
Debian GNU/Linux 13 webserver tty1
webserver login: frandyta
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

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Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
frandyta@webserver:~$ sudo ufw status
[sudo] password for frandyta:
Status: active

To Action From
--
WWW Full ALLOW Anywhere
OpenSSH ALLOW Anywhere
WWW Full (v6) ALLOW Anywhere (v6)
OpenSSH (v6) ALLOW Anywhere (v6)

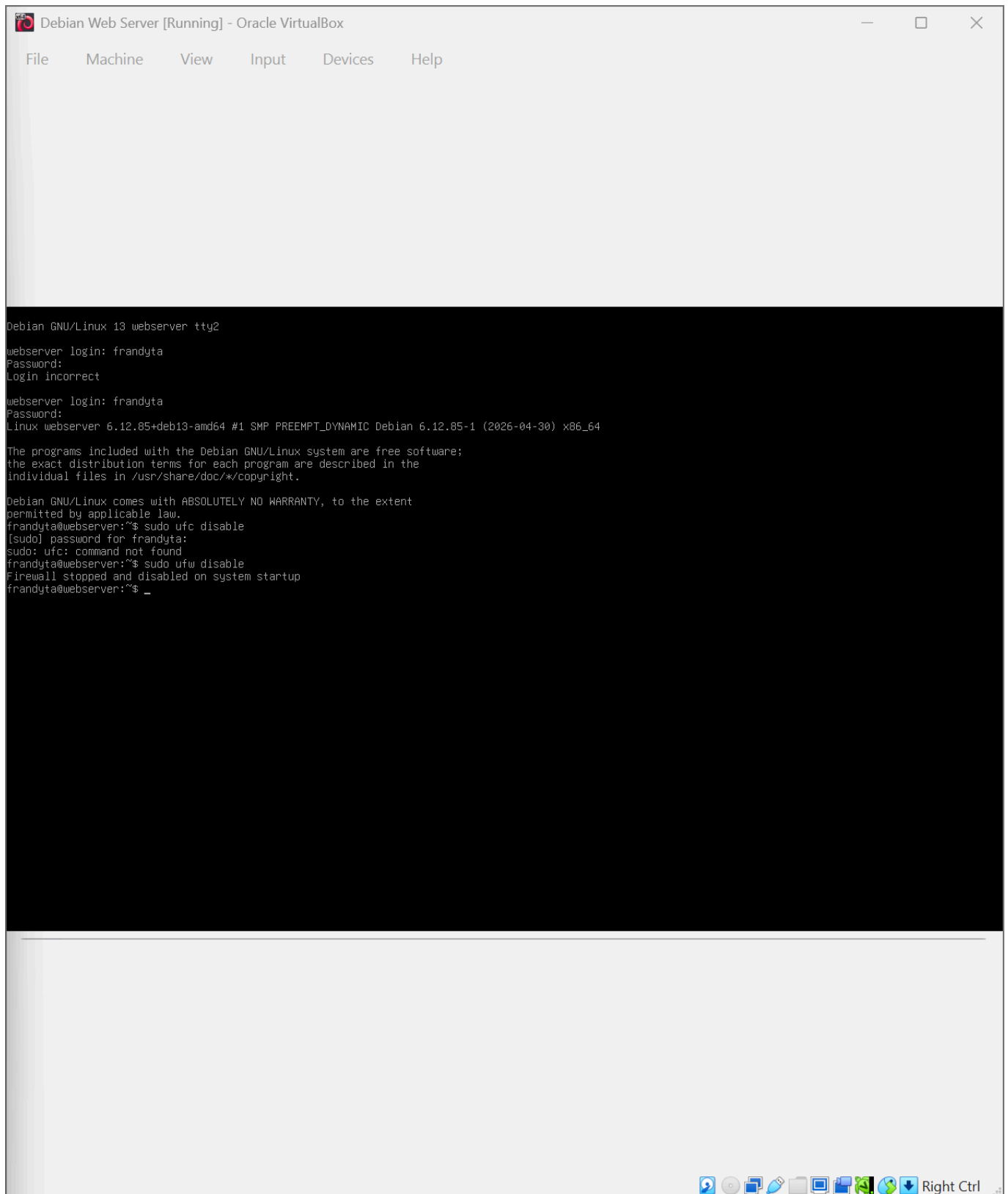
frandyta@webserver:~$
```

## Disable the Firewall

- The command is:

```
sudo ufw disable
```

- This command disables the firewall.

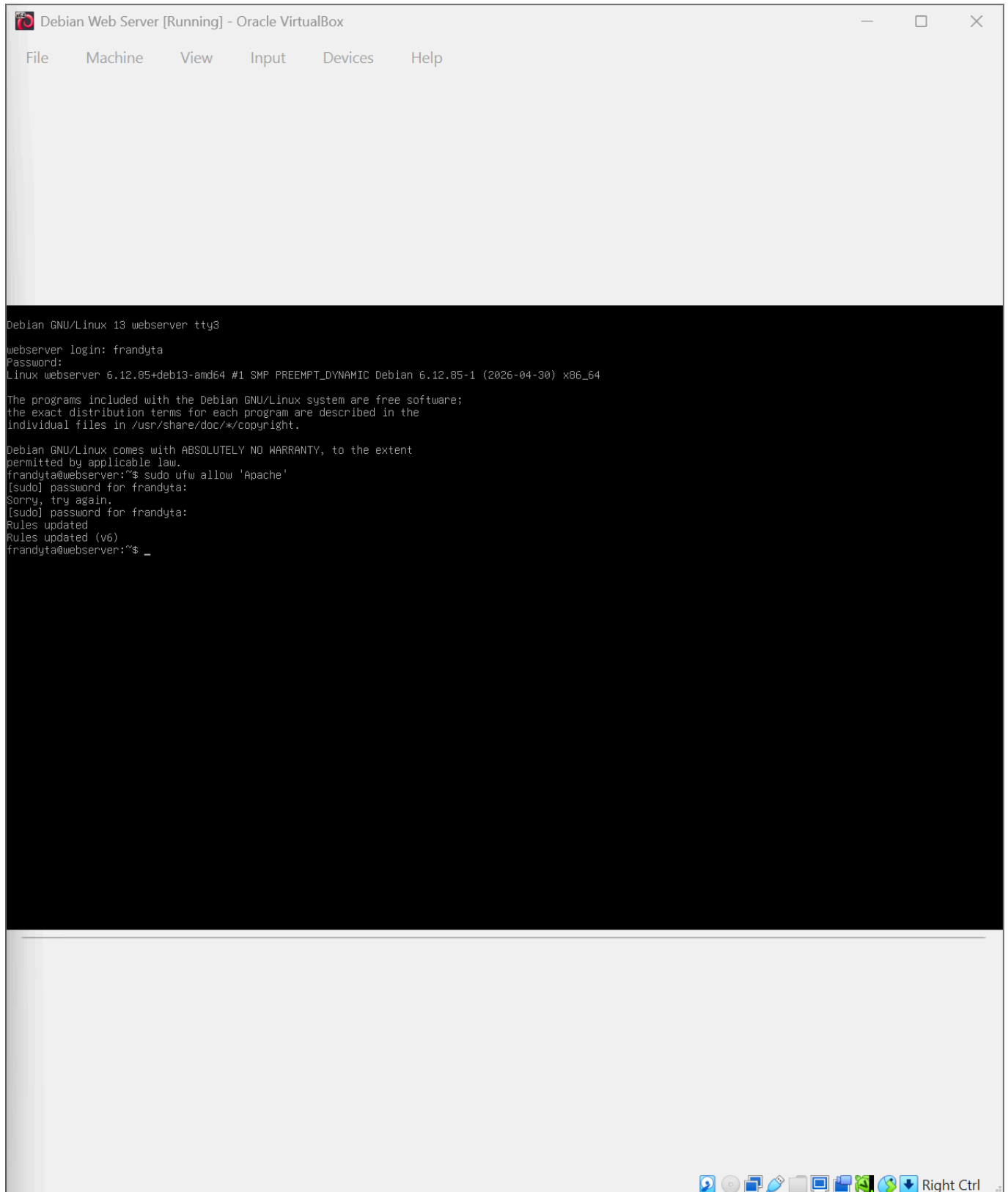


## Add Apache to the Firewall

- The command is:

```
sudo ufw allow 'Apache'
```

- This command allows Apache web traffic through the firewall.



```
Debian GNU/Linux 13 webserver tty3
webserver login: frandyta
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

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individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
frandyta@webserver:~$ sudo ufw allow 'Apache'
[sudo] password for frandyta:
Sorry, try again.
[sudo] password for frandyta:
Rules updated
Rules updated (v6)
frandyta@webserver:~$ _
```

## 5. What different commands do we use to work with Apache?

### 1. What is the command you use to check if Apache is running?

- The command is:

```
sudo systemctl status apache2
```



The screenshot shows a terminal window titled "Debian Web Server [Running] - Oracle VirtualBox". The terminal output for the command `sudo systemctl status apache2` is as follows:

```
frandyta@webserver:~$ sudo systemctl status apache2
[sudo] password for frandyta:
• apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 22:21:30 EDT; 10min ago
     Invocation: e84c21e46f7248f19cb8f9c1e70447e1
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 719 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 755 (apache2)
      Tasks: 55 (limit: 1093)
     Memory: 8.2M (peak: 8.5M)
        CPU: 548ms
    CGroup: /system.slice/apache2.service
            └─755 /usr/sbin/apache2 -k start
              └─757 /usr/sbin/apache2 -k start
                └─758 /usr/sbin/apache2 -k start

May 07 22:21:29 webserver systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 07 22:21:30 webserver apachectl[753]: AH00557: apache2: apr_sockaddr_info_get() failed for webserver
May 07 22:21:30 webserver apachectl[753]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this message
May 07 22:21:30 webserver systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-19/19 (END)
```

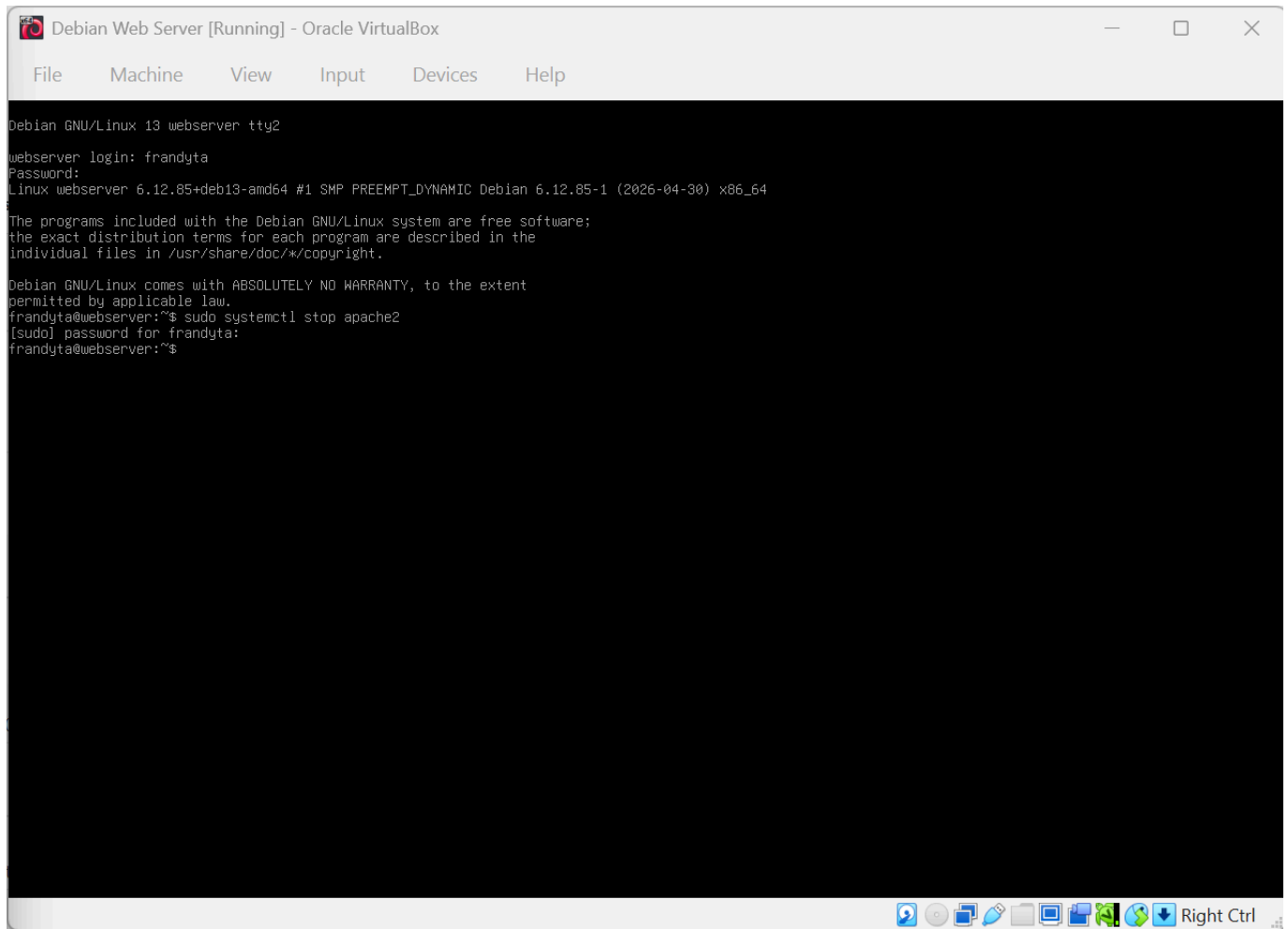
The terminal window includes a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The bottom status bar shows various system icons and the text "Right Ctrl".

## 2. What is the command you use to stop Apache?

- The command is:



```
sudo systemctl stop apache2
```



```
Debian GNU/Linux 13 webserver tty2
webserver login: frandyta
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

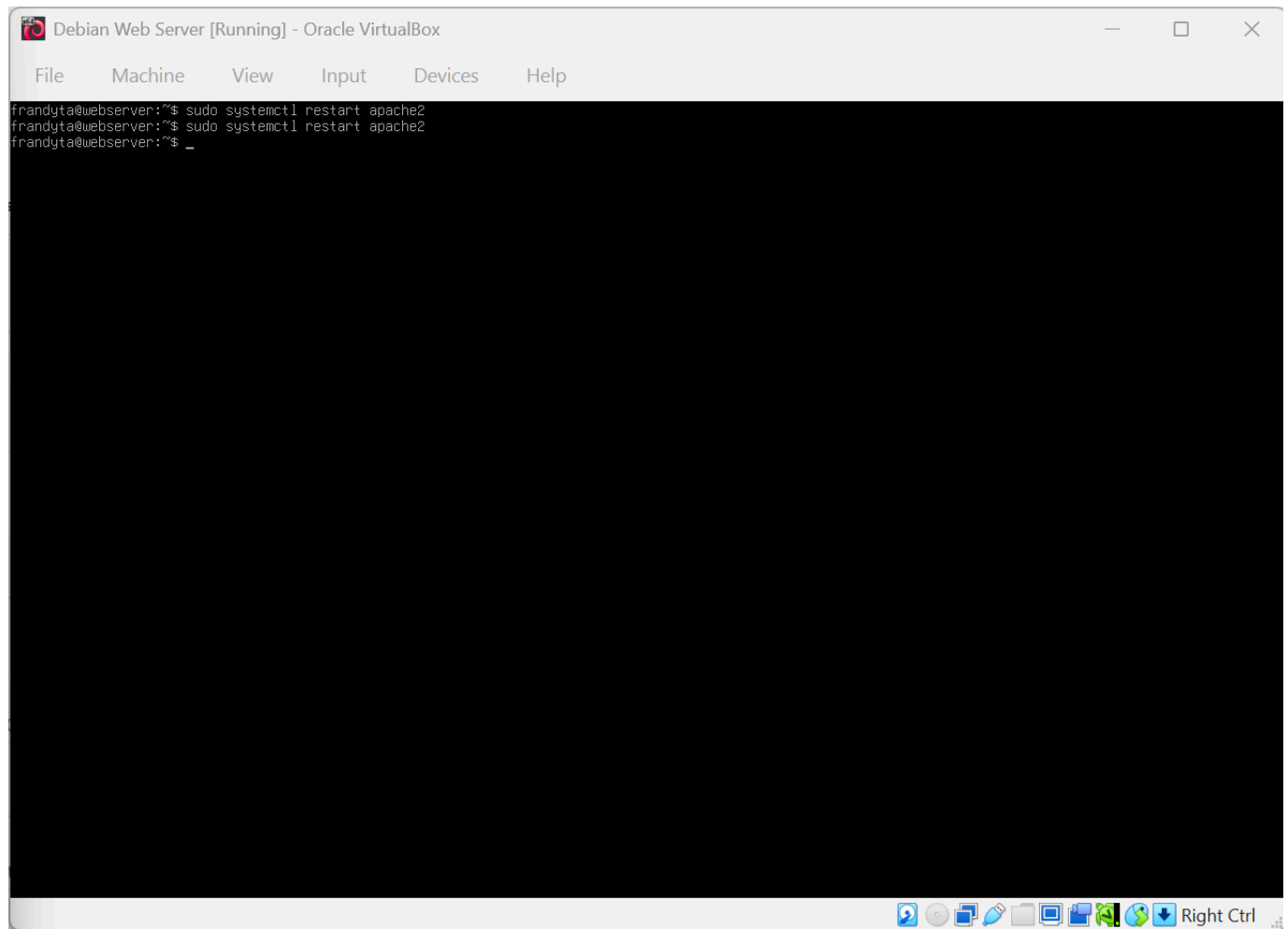
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permitted by applicable law.
frandyta@webserver:~$ sudo systemctl stop apache2
[sudo] password for frandyta:
frandyta@webserver:~$
```

### 3. What is the command you use to restart Apache?

- The command is:

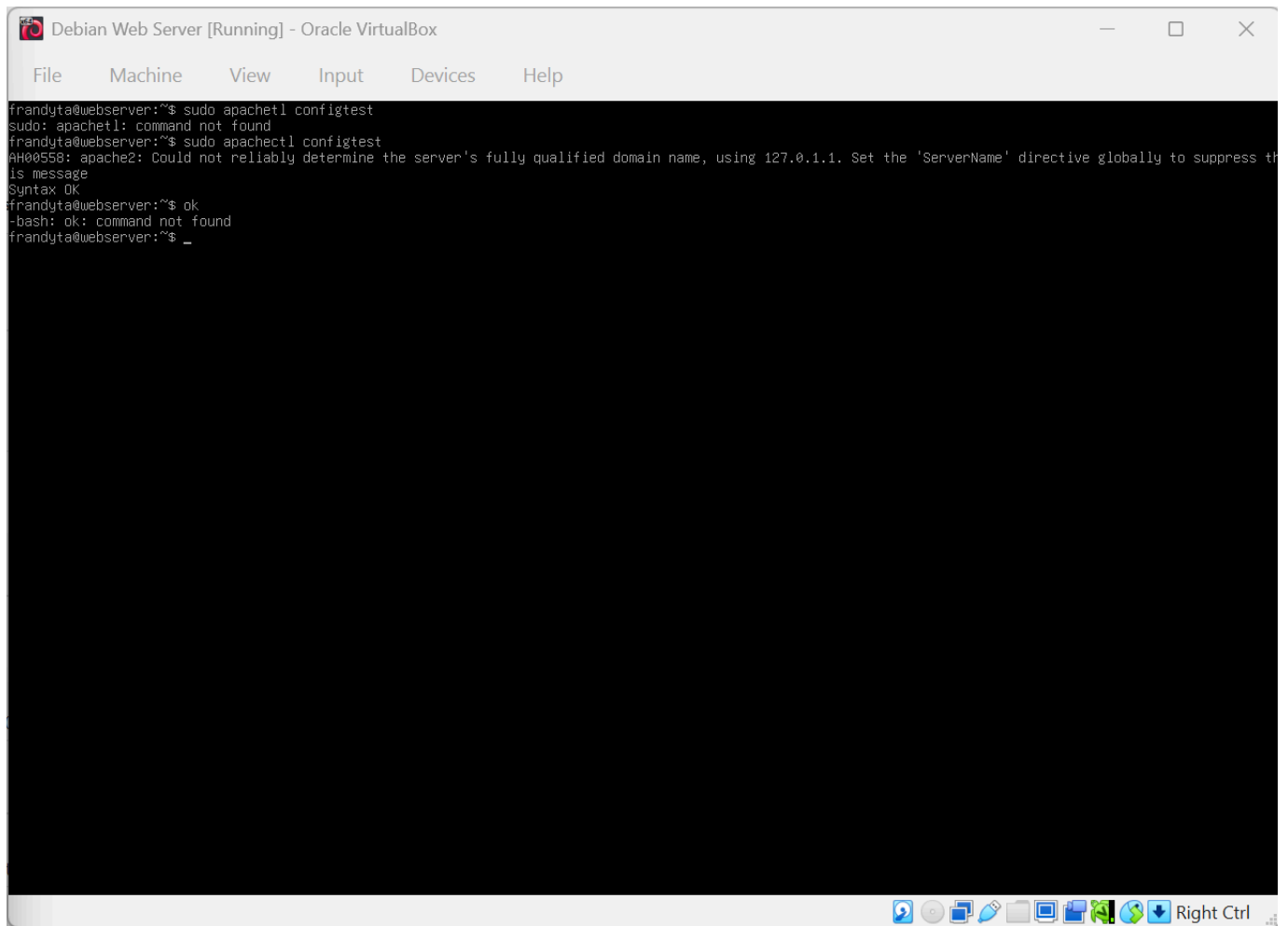
```
sudo systemctl restart apache2
```



#### 4. What is the command used to test Apache configuration?

- The command is:

```
sudo apachectl configtest
```

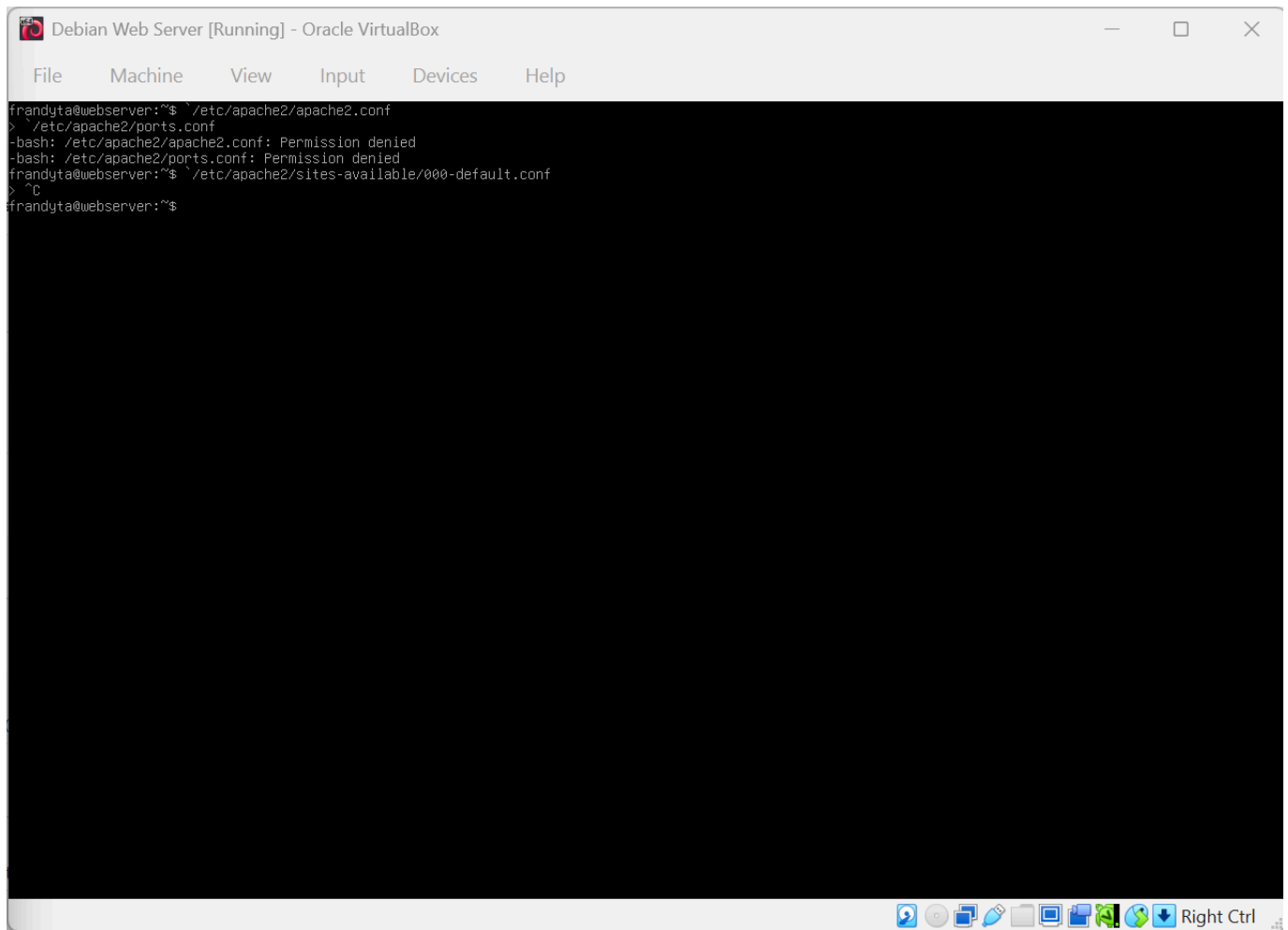


```
frandyta@webserver:~$ sudo apachetl configtest
sudo: apachetl: command not found
frandyta@webserver:~$ sudo apachectl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress th
is message
Syntax OK
frandyta@webserver:~$ ok
-bash: ok: command not found
frandyta@webserver:~$ _
```

5. What is the command used to check the installed version of Apache?

- The command is:

```
apache2 -v
```



```
Debian Web Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
frandyta@webserver:~$ nano /etc/apache2/apache2.conf
^C
-bash: /etc/apache2/apache2.conf: Permission denied
frandyta@webserver:~$ nano /etc/apache2/ports.conf
^C
-bash: /etc/apache2/ports.conf: Permission denied
frandyta@webserver:~$ nano /etc/apache2/sites-available/000-default.conf
^C
frandyta@webserver:~$
```

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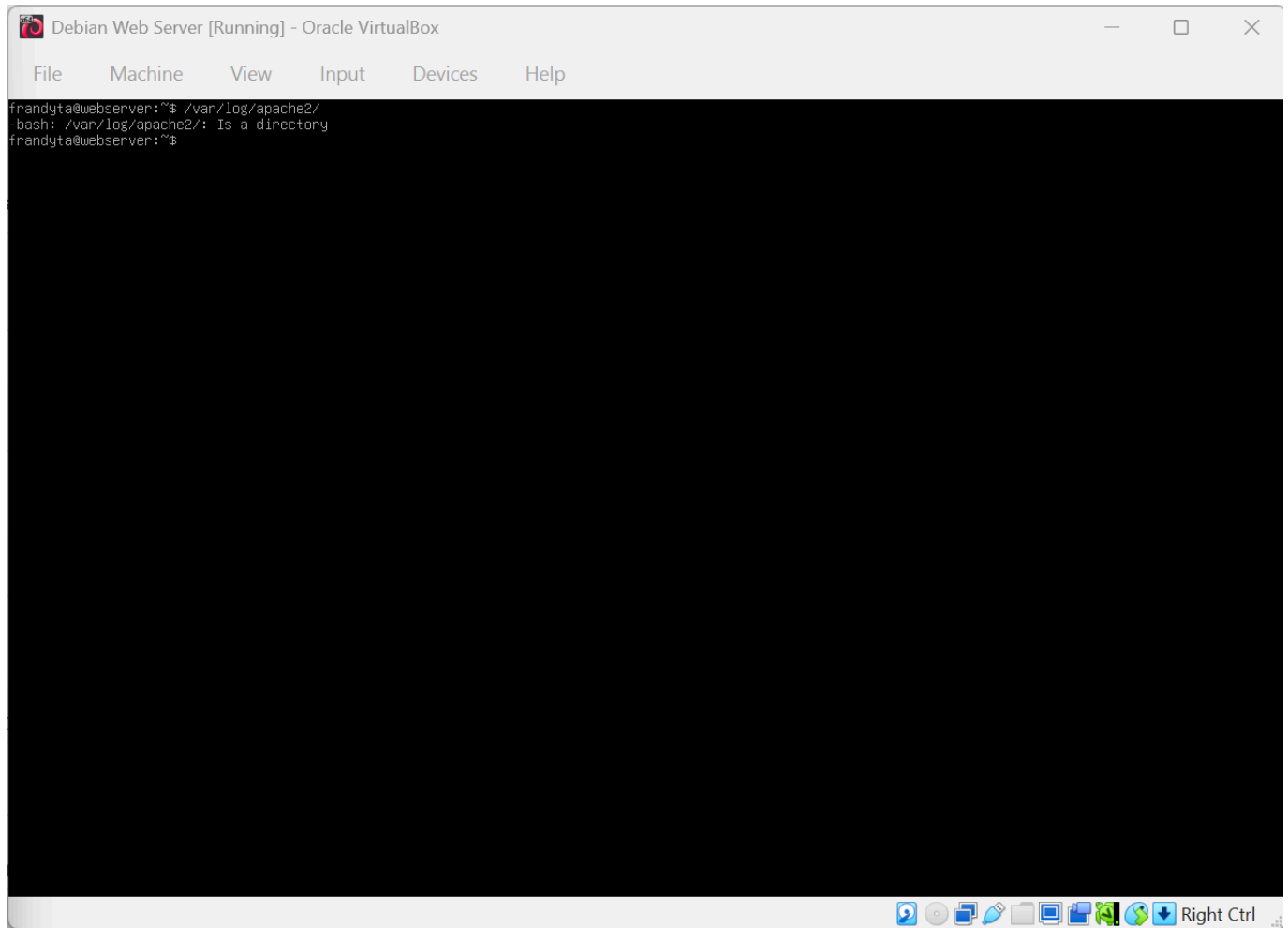
## 6. What are some common configuration files for Apache?

- `/etc/apache2/apache2.conf`
  - `/etc/apache2/ports.conf`
  - `/etc/apache2/sites-available/000-default.conf`
- 

## 7. Where does Apache store logs?

- Apache stores logs in:

```
/var/log/apache2/
```



8. What are some basic commands we can use to review logs?

View log file

```
cat /var/log/apache2/access.log
```

View last lines

```
tail /var/log/apache2/error.log
```

Monitor logs live

```
tail -f /var/log/apache2/access.log
```



reviewlogs